



PTT PHENOL COMPANY
LIMITED

PROCESS SPECIFICATION

PREFLASH COLUMN (V-2301)

Docu. No. : 14780-8120-PS-V2301
Project No. : 120117
Rev. No. : Z1
Date : MAY 11, 2016
Page : 1 / 9

TITLE :

PREFLASH COLUMN

(V-2301)

DISCIPLINE : **PROCESS**
DOCUMENT NO. : **14780-8120-PS-V2301**
PROJECT NAME : **PTT Phenol Train II Project**
PROJECT NO. : **120117**
OWNER : **PTT Phenol Company Limited (PPCL)**

Z1	MAY 11, 2016	AS BUILT	S.W.Park	J.S.Chung	-	S.W.Lee
F1	MAY 22, 2014	FINAL ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
A3	APR 18, 2014	APPROVED FOR DESIGN	S.W.Park	J.S.Chung	-	S.W.Lee
A2	DEC 11, 2013	APPROVED FOR DESIGN	S.W.Park	J.S.Chung	-	S.W.Lee
A1	OCT 2, 2013	APPROVED FOR DESIGN	S.W.Park	J.S.Chung	-	S.W.Lee
O1	JUN 27, 2013	COMMENT ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
REV. NO.	DATE	DESCRIPTION	PREP'N	REVIEW	QA REVIEW	APPROVAL
REV. NO.	PREPARATION DEPT.	PREPARATION	REVIEW	QA REVIEW	APPROVAL	
		S.W.Park	J.S.Chung	-	S.W.Lee	
Z1	PROCESS	MAY 11, 2016	MAY 11, 2016	-	MAY 11, 2016	

This Document is the property of POSCO Engineering Co., Ltd. Therefore, it shall not be released to Any third party without permission of an authorized personnel of the POSCO Engineering Co., Ltd.

[illegible]



Note: The information in this document is confidential and the property of UOP LLC and must not be disclosed to others or reproduced in any manner or used for any purpose whatsoever without its written permission.



VESSELS

PROJECT SPECIFICATION

963766 — 301

SHEET 3 / 9

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	29MAR12	JG	MYW				

General Notes

- A. Vessels shall be constructed in accordance with the following unless otherwise specified on the individual vessel specification sheet.

Standards:

UOP Standard Specification 3-11

UOP Standard Drawing 8-121

- B. Although the resulting inside diameter may be slightly different than specified, vessels without trays or packing, may be fabricated from pipe.
- C. The dimensioned location of some of the vessel level nozzles may be noted as min (minimum) or max (maximum). It is the responsibility of the contractor to adjust these dimensions to suit the specific level requirements specified on the UOP Piping and Instrument Diagram(s) and on UOP Standard Drawing 8-121. When additional nozzles are required for level instrumentation, the contractor shall specify them consistent with the size, class and facing requirements specified in the UOP Project Specification 301-Vessels and on the UOP Piping and Instrument Diagram(s).
- D. When system of measurement is metric, all dimensions are in millimeters unless noted otherwise.
- E. Contractor shall not locate TI thermowells in a downcomer. Contractor shall also ensure that TI thermowells are located to avoid interference with downcomers, weirs, distributors, collectors, baffles, chimneys, traps and all other vessel internals

Other specifications
referenced in this
specification:

UOP Standard Specification: 3-11, 3-15

UOP Standard Drawing: 3-115, 3-122, 3-123, 3-183, 3-190, 3-272, 3-278, 3-280, 3-305, 3-313, 3-322, 8-121

UOP Project Specification: 963766-307, 963766-401, 963766-808

R E V	VESSELS		
	SERVICE	PREFLASH COLUMN	ITEM NO V-2301
	Design	INT 3.5 kg/cm2 (g) @ 250°C	
	Conditions at Top	EXT FULL VACUUM @ 195°C	
	Operating Conditions	TOP 18.5 mmHg abs @ 52°C	
	Conditions	BTM 19.5 mmHg abs @ 70°C	
	Radiograph	MR	
	Postweld Heat Treat	MR	
	Material	SA 240 TYPE 304L (NOTE 1)	Corrosion Allowance 1.5
	This Vessel is in the Following Service(s):		
	Hydrogen	NO	
	Hydrofluoric (HF) Acid	NO	
	Hydrofluoric (HF) Acid (with Resid. Dem. Control)	NO	
	Wet Hydrogen Sulfide (H2S)	NO	
	Severe Cyclic	NO	
	Caustic	NO	
	Carbonate	NO	
	Amine	NO	
	Note:		
	Vessel Heads		
	Top	2:1 ELLIPTICAL	
	Bottom	2:1 ELLIPTICAL	
	Accessories Applied by Fabricator		
	Ladder & Platform Clips	AS REQUIRED	
	Insulation Clips & Rings H (50)	YES	
	Vessel Support	YES	
	Nozzles and Manways		
	Mark No	Size Inches	Service
	AB 1	4	VENT
	AH 2	2	PI
	AI 1	3	INLET (REFLUX)
	AJ 1	3	EQUALIZING
	AO 2	72	VAPOR OUTLET (7)
	AU 4	12	RELIEF NOTE 4
	AZ 1	24 ID	MANWAY
	BB 1	24 ID	MANWAY
	BC 2	1120 mm ID	EXCHANGER, NOTES 3,5,10,11
	BD 1	52	INLET
	CB 1	24 ID	MANWAY
	CD 1	1160 mm ID	EXCHANGER, NOTES 5,6,10,11
	CF 3	2	TSH, NOTE 15 (2)
	CG 1	3	QUENCH
	CH 1	6	QUENCH
	CL 1	2	LSH, NOTE 14 (2)
	CM 2	2	DRAIN, NOTES 2&13 (2)
	CP 2	2	DRAIN RETURN, NOTE 14 (2)
	CR 2	3	TUB LEVEL (2)
	CS 2	3	TUB LEVEL (2)
	CX 2	2	PI (2)
	Class - ASME CL150		
	(2)- ASME CL300 (7)- SPECIAL		
	Facing - RAISED FACE		
	Normal Liquid Level = EMPTY		
	Specific Gravity = 0.918		
	Delta-P Packing (Total) = 1.1 mm Hg		

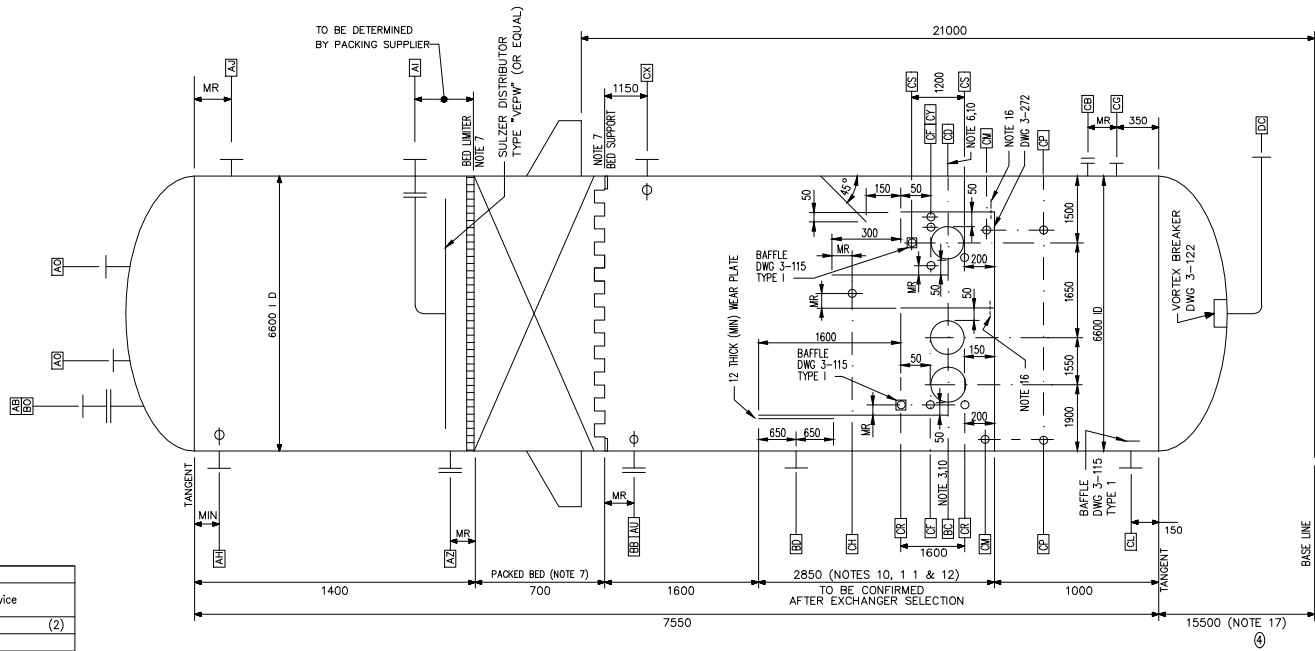
NOTES:

- VESSEL SHALL BE CONSTRUCTED IN ACCORDANCE WITH UOP STANDARD SPECIFICATION 3-15. ALTERNATELY, ENTIRE VESSEL SHALL BE CARBON STEEL LINED WITH 3 mm THICK STAINLESS STEEL SA-240 TYPE 304L. ALL INTERNALS TO BE OF SA-240 TYPE 304L. ALL NOZZLES AND MANWAYS TO BE LINED AND FACED WITH SA-240 TYPE 304L, AND LINING IS CORROSION ALLOWANCE. CARBON STEEL VESSEL SHALL BE FABRICATED IN ACCORDANCE WITH UOP STANDARD SPECIFICATION 3-1 1.
- ENTIRE COLUMN MUST FREE DRAIN COMPLETELY.
- A U-TUBE HEAT EXCHANGER BUNDLE WILL BE INSERTED HORIZONTALLY INTO EACH OF THE COLUMN NOZZLES "BC". SEE UOP PROJECT SPECIFICATION - 401 FOR INFORMATION ON THE EXCHANGER (ITEM E-2302 A/B).
- RELIEF NOZZLES TO BE LOCATED AT SAME ELEVATION.
- MINIMUM TO ACCOMMODATE EXCHANGER.
- A U-TUBE HEAT EXCHANGER BUNDLE WILL BE INSERTED HORIZONTALLY INTO THE COLUMN NOZZLE "CD". SEE UOP PROJECT SPECIFICATION - 401 FOR INFORMATION ON THE EXCHANGER (ITEM E-2303).
- FOR PACKED BED, DISTRIBUTOR, BED LIMITER AND BED SUPPORT SEE UOP PROJECT SPECIFICATION 307 (TRAYS AND PACKING).
- DIAMETER OF STABBED-IN REBOILER NOZZLE IS TO BE CONFIRMED AFTER SELECTION OF REBOILER. VESSEL FLANGE MATCHING REBOILER FLANGE TO BE FURNISHED BY REBOILER SUPPLIER AND INSTALLED BY VESSEL FABRICATOR.
- DIMENSIONS OF SURROUNDING NOZZLES TO BE VERIFIED AFTER SELECTION OF REBOILER.
- BAFFLES ARE TO BE REMOVABLE AND GASKETED, IF REQUIRED. MR DESIGN.
- INVERT OF NOZZLES "CM" TO BE FLUSH WITH BOTTOM OF TUB.
- LOCATE NOZZLES "CL" AND "CP" BELOW REBOILER TUB FLOOR.
- "CF" NOZZLES TO BE LOCATED IN LIQUID.
- PROVIDE 2 DRAIN HOLES, 6mm DIA IN BOTTOM OF EACH REBOILER OVERFLOW BAFFLE, LOCATED OPPOSITE FROM EACH OTHER AND AS CLOSE AS POSSIBLE TO VESSEL WALL TO ALLOW COMPLETE DRAINAGE OF REBOILER TUB.
- SKIRT HEIGHT CAN BE REDUCED IF COLUMN IS SUPPORTED IN STRUCTURE.

PROJECT SPECIFICATION

963766 - 301 SHEET 4 / 9

REV	DATE	BY	APP'D	REV	DATE	BY	APP'D
0	29MAR12	TS	MYW	4	13SEP12	TS	MYW
1	06APR12	JG	MYW				
2	11MAY12	JG	MYW				
3	13AUG12	JG	MYW				



Drawings Referred to in this Specification					
3-115	3-122	3-272			

Revision
Indication



PROJECT SPECIFICATION

963766 — **307**

SHEET 5 / 9



**TRAYS AND PACKING
(RANDOM AND STRUCTURED)**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	02Apr12	TS	MYW				

**UOP Phenol Process Unit
Concentration, Decomposition and Neutralization Section**

Item Number	Service	Sheet
	General Notes	6 & 7
V-2301	Preflash Column	8
	Packed Bed Data	9

Other specifications
referenced in this
specification:

UOP Standard Specification: 3-18

UOP Standard Drawing: 3-280, 3-282

UOP Project Specification: 963766-301

Note: The information in this document is confidential and the property of UOP LLC and must not be disclosed to others or reproduced in any manner or used for any purpose whatsoever without its written permission.



TRAYS AND PACKING (RANDOM AND STRUCTURED)

PROJECT SPECIFICATION

963766 — 307

SHEET 6 / 9

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	02Apr12	TS	MYW				

The following general notes apply to all items listed on individual specification sheet:

- A. All items shall comply with the requirements, terms and conditions of this Project Specification and UOP Standard Specification 3-18. For details of vessels see UOP Project Specification 301.
- B. All trays are numbered from **TOP #1 TO BOTTOM**.
- C. Abbreviations: CS = carbon steel; Type 410 = 13% chromium alloy; Type 304 = 18% chromium -8% nickel alloy; Type 316 = 16% chromium – 12% nickel alloy; "M.R." or "MR" = manufacturer's and / or contractor's responsibility. SS may be any of the following – Type 410, Type 304 or Type 316. See UOP Standard Specification 3-18 for additional clarification on materials.
- D. When system of measurement is metric, all dimensions are in millimeters except as noted.
- E. For vessels where the inside diameter is indicated to be minimum (min), adjustments in tray dimensions to accommodate the final inside diameter selected shall be to the chord height of side downcomers which are also indicated as minimum (min).
- F. Vessel nozzle and internals orientation errors are a common source of problems for trayed columns. UOP does not supply specific orientation drawings in order to allow flexibility to the contractor, the vessel fabricator and the internals supplier. The contractor is to ensure sufficient communication between the various parties to avoid orientation difficulties. Some common orientation restrictions follow.
 - Thermowells that are inserted into trayed distillation columns must not contact internals such as support rings or downcomer panels.
 - Reboiler return nozzles should be oriented to minimize vapor interaction with the liquid from the bottom tray traps.
 - Adjacent distillation column trayed sections normally have specific orientation requirements to facilitate vapor and liquid movement between the sections. The supplier of the vessel internals (trays and packing) must provide the vessel fabricator with orientation drawings.
 - Nozzles shall not be located in downcomers.

The following notes apply only when listed for individual item of specification sheet:

1. One pass.
2. Two pass, center downcomers.
- 2x. Two pass, side downcomers.
3. Three pass.
4. Four pass, intermediate downcomers.
- 4x. Four pass, side and center downcomers.
5. Tray(s) shall be liquid tight by seal welding, except at manway(s) where gasketing shall be used.
6. Tray(s) including downcomer(s) shall be liquid tight by seal welding, except at manway(s) where gasketing shall be used.



**TRAYS AND PACKING
(RANDOM AND STRUCTURED)**

PROJECT SPECIFICATION

963766 — **307**

SHEET 7 / 9

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	02Apr12	TS	MYW				

7. Do not provide drain holes. Tray(s) shall be liquid tight.
8. Stiffeners and support members transverse to the liquid flow direction shall not exceed 2" (50 mm) in depth and shall not be located above upcomers, or within a lateral distance from upcomers equal to one tray space.
9. Trays shall be assembled on rods and spacers or arranged in cartridges with trap attached to downcomer of bottom tray. Furnish Teflon or metal wall wiper at each tray for liquid seal.
10. All tray assemblies shall be designed for installation and removal from top and bottom.
11. Each tray shall have froth initiator(s) in accordance with UOP Standard Drawing No 3-130.
12. Bubble caps shall be in accordance with UOP Standard Drawing No _____ Type No _____.
13. Ceramic Fiber gasketing shall be used as required for proper design and installation of tray(s), manway(s) and / or distributors for packed beds.
14. Each tray shall have a total of ___ - ___ diameter holes on approximately ___ square, equilateral triangular pitch.
15. Each tray shall be designed for an upward pressure differential of ___.
16. Drawoff well(s) on tray No ___ shall be ___ deep and a minimum of ___ wide and extend under entire downcomer area.
17. Provide an additional ___ percent of the required random packing to cover losses.
18. Minimum thickness of tray components is 12 gauge (0.104" / 2.6 mm). A corrosion allowance is not required.
19. Bolts and nuts for trays shall be carbon steel (ASTM A307 Grade A for bolts and A563 Grade A for nuts).
20. Tray No(s) ___ shall be designed for a net uplift of 100 pounds per square foot (488 kilograms per square meter).
Tray No(s) ___ shall be designed for a total downward load of 100 pounds per square foot (488 kilograms per square meter) plus dead weight.
21. Customer/Contractor must verify that the structural and mechanical condition of the existing tray(s) and/or tray supports is satisfactory for continuing operation.
22. UOP MD and Slotted Sieve tray decks, downcomers, installation hardware and distributors; and Raschig Super-Rings and distributors are to be supplied by:

Supplier (Mandatory):

UOP LLC
175 East Park Drive
PO Box 986
Tonawanda, NY 14151-0986
Telephone 716-879-7615
Facsimile 716-879-7215

Revision
Indication



PROJECT SPECIFICATION

963766 — **307**

SHEET 8 / 9



**TRAYS AND PACKING
(RANDOM AND STRUCTURED)**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	02Apr12	TS	MYW				

Service: Preflash Column

Item Number: V-2301

Item No	Quan Assem	ID of Vessel	Material	Description	Dwg or Sht No	Dimensions	Tray No	Notes (Sheet 6&7)
1	1	6600	Type 304	Distributor	5	(Note 1)	Above Bed	
2	1	6600	Type 304	Bed Limiter	5		Above Bed	
3	-----	6600	Type 304	Packing	5		Bed	
4	1	6600	Type 304	Bed Support	5		Below Bed	

Note:

1. Distributor to be Sulzer Type “VEPW” (or equal).

Tray sections shall pass through a 24” ID vessel manway.

Note: The information in this document is confidential and the property of UOP LLC and must not be disclosed to others or reproduced in any manner or used for any purpose whatsoever without its written permission.



**TRAYS AND PACKING
(RANDOM AND STRUCTURED)**

PROJECT SPECIFICATION

963766 — **307**

SHEET 9 / 9

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	02Apr12	TS	MYW				

Service: Preflash Column

Item Number: V-2301

Packed Bed Data

Bed Identification	Single Bed Design Case			
Vessel ID, mm	6600			
Bed Depth				
Structured Packing, mm	700			
Random Packing, mm				
Packing Type (or equal)				
Structured	Sulzer Mellapak 250X			
Random				
Structural Design Temperature, °C	250			
Fluid Description	Cumene, CHP			
Foam Derating Factor	1.0			
Top of Bed				
Pressure, mmHg abs	18.5			
Vapor-Flow Rate, kg/h	138376			
Density, kg/m ³	0.109			
Viscosity, cP	0.0068			
Temperature °C	53			
Liquid-Flow Rate, kg/h	19055			
Density, kg/m ³	839			
Viscosity, cP	0.557			
Surface Tension, dynes/cm	25.3			
Bottom of Bed	Top loads govern			
Pressure, mmHg abs	19.5			
Vapor-Flow Rate, kg/h	132923			
Density, kg/m ³	0.111			
Viscosity, cP	0.0070			
Temperature °C	64			
Liquid-Flow Rate, kg/h	13908			
Density, kg/m ³	850			
Viscosity, cP	0.655			
Surface Tension, dynes/cm	25.2			
Temperature °C	53			
External Feed Above Section				
Distributor Type (or equal)	By packing vendor			
Temperature °C	39			
Vapor-Flow Rate, kg/h	-			
Density, kg/m ³	-			
Viscosity, cP	-			
Liquid-Rate, kg/h	19391			
Density, kg/m ³	847			
Viscosity, cP	0.622			
Surface Tension, dynes/cm	26.3			

Note: The information in this document is confidential and the property of UOP LLC and must not be disclosed to others or reproduced in any manner or used for any purpose whatsoever without its written permission.